

Smart Fuel Optimization & Resource Management System

Project Domain / Category

Web Application

Abstract / Introduction

In developing economies like Pakistan, fuel is a very essential but scarce commodity that is highly dependent on imports. When the supply chain is disrupted during times of crisis like economic instability, war conditions or any other factors that activate such crisis, the problem of fuel shortage can severely affect transportation, logistics and daily life. People and small organizations do not have mechanisms to control the consumption of fuel, which leads to the waste of fuel and financial burden.

This project proposes the construction of scalable web-based Smart Fuel Optimization and Resource Management System which will allow users to monitor, plan and optimize fuel consumption. The system will be inspired by such systems as Google Maps (to optimize the route), Mint (to gain an understanding of the budget), and Zoho Inventory (to track the resources), and will comprise a hybrid platform that will combine fuel tracking, smart trip planning, and consumption analytics.

The system will give the users some recommendations based on the data, including an identification of the fuel efficient routes, unnecessary travelling reduction, and effective fuel budget management. It will also feature a crisis mode, which will allow users to focus on the most important tasks and extend fuel availability during shortages. It will be available on web browsers and will be scalable, thus can serve individuals as well as small organizations.

Functional Requirements:

- ✚ The system shall allow users (individual, organizational, and admin) to register an account, log in securely, update their personal profile, and log out.
- ✚ The system shall allow users to add new vehicles with attributes including fuel efficiency and tank capacity.
- ✚ The system shall allow users to view, update, or delete vehicle information at any time.
- ✚ The system shall allow users to create, read, update, and delete fuel purchase records linked to vehicles, including date, quantity, and cost.
- ✚ The system shall allow users to create trips by specifying a starting point and destination.
- ✚ The system shall calculate the estimated fuel required and suggest optimized routes to minimize fuel consumption.
- ✚ The system shall provide a “crisis mode” that recommends strategies such as combining trips or avoiding non-essential travel to conserve fuel.
- ✚ The system shall allow users to set monthly fuel budgets, track fuel expenses automatically, and display alerts when budgets are exceeded through a user dashboard.
- ✚ The system shall provide visual charts and graphs showing total fuel consumption, costs over time, and per-vehicle statistics.

- ✚ The system shall allow users to filter and interact with dashboards to analyze fuel usage patterns.
- ✚ The system shall display planned and optimized routes on a map, including distance, estimated fuel consumption, and travel time.
- ✚ The system shall allow users to zoom, pan, and interact with the map for better trip planning.
- ✚ The system shall notify users of critical events, including budget limits exceeded, inefficient trips, or recommended actions during crisis mode.

Tools:

Development Technologies:

Front-end:

HTML5/CSS3/JavaScript/React.js

Back-end:

Node.js, Express.js

Database: MongoDB

Development Tools:

Visual Studio Code

Figma (for UI/UX design)

Postman (API testing)

GitHub (version control)

Supervisor:

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